

PUBLIC EXHIBIT 17

Meet Oura

Discover the Brand-New Oura Ring 4



Author: Oura Team | October 3, 2024



Oura Ring 4 is our most personal, accurate, and comfortable smart ring yet.

With scientifically validated sensing capabilities, Oura Ring 4 is built around [Smart Sensing](#), a breakthrough technology that adapts to your unique physiology to deliver accurate, continuous data.

Featuring recessed sensors and a fully titanium inner and outer coating, Oura Ring 4 is sleeker and more comfortable than ever to wear all day long, from workout to night out. In keeping with our mission to make health a daily practice for *every* body, Oura Ring 4 is also available in a wider range of sizes—now extending from size 4 to size 15.

Sleeker. Smarter. Made for you.

[Shop Oura Ring 4](#)

Furthermore, all Oura members with an Oura Ring Gen3 or Oura Ring 4 will experience upgrades to several features, including Daytime Stress, Automatic Activity Detection, and Cycle Insights, as well as a newly [redesigned Oura App](#).

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“Oura Ring is designed to be a personal health companion, giving our members a holistic view of their health by pairing personal data with meaningful, actionable insights to facilitate long-term changes,” says Tom Hale, Oura CEO. “The next-generation Oura Ring 4, reimagined Oura App, and the development of the Smart Sensing platform represent significant steps forward for wearables across accuracy, comfort, personalization, and design.”

Oura Ring 4 starts at \$349, with Oura Membership pricing at \$5.99 per month or \$69.99 per year.

Keep reading to learn more.

Oura Ring 4: More Accurate Than Ever Before



Crafted with precision, the [accuracy of Oura Ring](#) has been validated by research institutions across the globe, such as the University of Tokyo and the [National University of Singapore](#). In Oura Ring 4, the new Smart Sensing technology provides an unparalleled degree of accuracy and personalization.

This breakthrough health sensing platform works with the ring’s research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

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This breakthrough health sensing platform works with the ring's research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

LEARN MORE: [Smart Sensing: Unparalleled Accuracy and Personalization in Oura Ring 4](#)

A Wearable That's Truly Wearable, Day and Night



Oura Ring 4 is sleeker and more comfortable than its predecessors, allowing it to blend seamlessly into your lifestyle. With a fully titanium coating inside and out, it's gentle to the touch, even on sensitive skin, and slips on and off your finger with ease.

Plus, the powerful health sensors are now recessed, reducing friction and pressure on your fingers when shaking hands or gripping objects. With the lower profile and advancements in comfort of Oura Ring 4, you might even forget you're wearing it—until the compliments start coming.

Choose from Expanded Sizes & Stunning Finishes



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Find the right fit for you *and* your personal style. Oura Ring 4 comes in an expanded range of sizes (4-15), the widest size range available on the smart ring market. Select from six gorgeous finishes, including a new, deep black that's more durable than before.

LEARN MORE: [Inside the Ring: The Making of Oura Ring 4](#)

Enjoy Up to 8 Days of Battery Life



Never miss a beat. Oura Ring 4 delivers up to eight days of battery life on just a single charge, and fully charges in up to 120 minutes, depending on current battery

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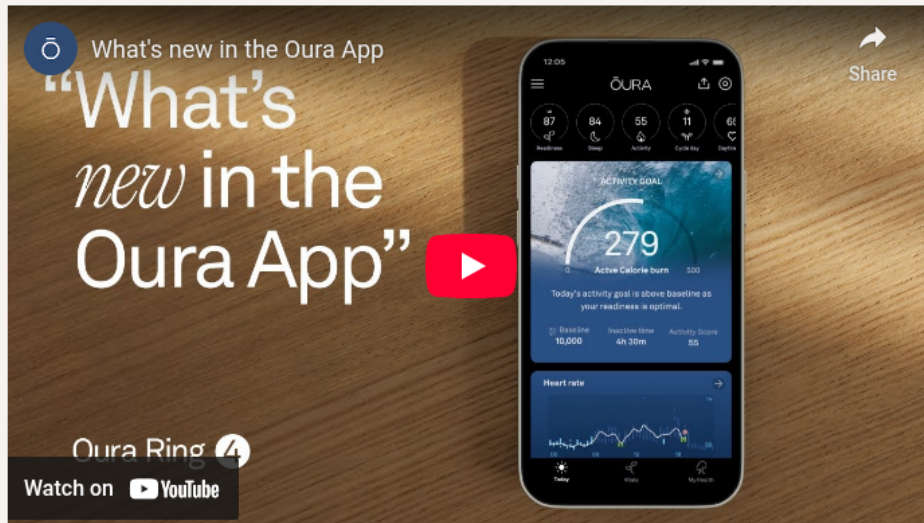
the Year!

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Never miss a beat. Oura Ring 4 delivers up to eight days of battery life on just a single charge, and fully charges in up to 120 minutes, depending on current battery level.*

Plus, the charger itself has been redesigned, with an aluminum body and grippy mat on the bottom that provides more stability and improved anchoring as it charges your ring.

Experience a Redesigned Oura App



Currently rolling out to all members, the award-winning Oura App has been redesigned to bring your short- and long-term health into focus, with three new, streamlined tabs: Today, Vitals, and My Health. This easier-to-navigate setup provides an at-a-glance view into your holistic wellness data on a daily basis and trends over time.

READ MORE: [Introducing the New Oura App Design](#)

New Feature Upgrades





Over time, we’ve significantly invested in improving our software experience. As of October 3, 2024, new software updates will begin rolling out to both Oura Ring 4 and Gen3 members.

Automatic Activity Detection (AAD) now includes **average heart rate and heart rate zones for 40+ activities**, automatically giving you a more comprehensive view of your movement throughout the day.

Cycle Insights now includes **Fertile Window**, a new feature that gives women who are trying to conceive even more information about their cycle and their fertility journey.

Daytime Stress now provides more context about the relationship between your daytime stress levels and your tags, movement, and activities. This will help you understand how your daily habits affect your stress and your recovery.

Oura Ring 4 vs. Oura Ring Gen3

How does Oura Ring 4 compare to Oura Ring Gen3? The comparison chart below provides specifics, such as the difference in material and design, the sensing platform, and sizing.

In addition, Oura Ring 4 is now capable of storing up to two weeks of data, depending on the type and frequency of use.

Compare Oura Ring Models

Style	 Oura Ring 4	 Oura Ring Gen3
Design & Material	Fully titanium exterior & interior Recessed interior sensors at 0.3mm	Titanium exterior, epoxy interior Raised interior domes (sensors) at 1.3mm
Sensing Platform	Accuracy is powered by Smart Sensing that dynamically selects the optimal signal path	Measurements powered by a fixed, static configuration of optical paths
Battery	Up to 8 days	Up to 7 days
Sizing	Sizes 4-15	Sizes 6-13

High Blood Pressure in Pregnancy: What You Need to Know

6 Lifestyle Habits to Manage Blood Pressure Naturally

What Is Hypertension? Spot the Signs, Lower Your Risk & Protect Your Heart



Design & Material	Fully titanium exterior & interior	Titanium exterior, epoxy interior
	Recessed interior sensors at 0.3mm	Raised interior domes (sensors) at 1.3mm
Sensing Platform	Accuracy is powered by Smart Sensing that dynamically selects the optimal signal path	Measurements powered by a fixed, static configuration of optical paths
Battery	Up to 8 days	Up to 7 days
Sizing	Sizes 4-15	Sizes 6-13
Membership Pricing	\$5.99/month	\$5.99/month

All battery claims depend on **Oura Ring size, **Oura App settings**, **feature configuration**, **activity**, **age of battery**, **usage**, and **many other factors**; actual results will vary. Battery life and charge cycles vary by use and settings. Battery tests were conducted across various sizes and scenarios. See [Ring Battery tips](#) for more information.*

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HEALTH
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Oura Ring 4

Oura Ring 4 is the next evolution of the revolutionary smart ring, designed to help you get up close and personal with your health. Oura Ring 4 is built around a breakthrough sensing technology, Smart Sensing, which adapts to your finger to deliver accurate, continuous data, day and night.

- Oura Ring 4 is available through the [Oura online store](#) and various [authorized retailers](#)
- Oura Ring 4 Ceramic:
 - All colors (Midnight, Cloud, Petal, and Tide) are available through the [Oura online store](#) and [Best Buy US](#) and [Best Buy Canada](#)
 - Cloud, Petal, and Tide are available at various [authorized retailers](#)

If you're setting up your Oura Ring 4 as a new or current member, see the instructions in [Set Up an Oura Ring](#).

Also see [Oura Ring 4 FAQs](#).

[Hardware](#)

[Software](#)

[Oura Ring 4 vs. Oura Ring 4 Ceramic](#)

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Hardware

Oura Ring 4 features a fully titanium design and recessed sensors on the interior, enhancing both durability and wearability. Oura Ring 4 is available in six finishes: Black, Silver, Stealth, Brushed Silver, Gold, and Rose Gold.

Oura Ring 4 Ceramic is made from high-performance zirconia ceramic, a premium material selected for its scratch resistance, luster, and color. Oura Ring 4 Ceramic is available in four colors: Midnight, Cloud, Petal, and Tide.

All Oura Ring 4 models are available in sizes 4-15.

The sizing for Oura Ring 4 is slightly different from Oura Ring Gen3. Make sure to use the right sizing kit for the most accurate fit.

- Oura Ring 4 Sizing Kit: white ring sizers, sizes 4–15
- Oura Ring Gen3 Sizing Kit: black ring sizers, sizes 6–13

For the best fit and most accurate readings, align your ring so that the sensors are against the palm side of your finger.



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Sensors

- Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life
- Red and infrared LEDs measure blood oxygen while you sleep
- Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep
- Digital temperature sensor measures average body temperature variations
- Accelerometer tracks movement and activity 24/7

External Surface Materials

- Oura Ring 4:
 - Titanium with physical vapor deposition (PVD) coating in Black, Silver, Brushed Silver, Gold, and Rose Gold
 - Titanium with Diamond-like Carbon (DLC) coating in Stealth
- Oura Ring 4 Ceramic:
 - Zirconia ceramic

Inner Surface Materials

- Non-allergenic, titanium inner shell. This is a medical grade, BPA-free inner surface
- Shock resistance and bonding between printed circuit board and ring outer frame

Connectivity

- Bluetooth Low Energy (Bluetooth Smart®)
- EMF-Safe and allows Airplane Mode
- FCC certified

Other Specifications

- Battery: Typically 5-8 days. Battery life and longevity depend on Oura Ring size, Oura App settings, feature configuration, activity, battery age, usage, and many other factors; actual results will vary. Battery life and charge cycles vary by use and settings. Battery tests were conducted across various sizes and scenarios. See Oura Ring 4 Battery Life and Ring Battery Tips for more information
- Charging time: Typically between 20-80 minutes depending on the ring starting charge level
- Water Resistance: 100m/328ft, suitable for sauna and water sports but not



- cycles vary by use and settings. Battery tests were conducted across various sizes and scenarios. See [Oura Ring 4 Battery Life](#) and [Ring Battery Tips](#) for more information
- **Charging time:** Typically between 20-80 minutes depending on the ring starting charge level
 - **Water Resistance:** 100m/328ft, suitable for sauna and water sports but not diving
 - **Operating Temperature:** -10–54 °C / 14–129 °F. Charging at room temperature recommended

See more information in our [Product Safety & Use article](#).

Software

Oura Ring 4 with active membership includes:

- [Detailed sleep analysis](#)
- [24/7 heart rate tracking \(Daytime, Nighttime, Activity\)](#)
- [Integration with third-party health and wellness apps like Strava](#)
- [Advanced temperature monitoring](#)
- [Blood Oxygen Sensing \(SpO2\)](#)
- [Daytime Stress](#)
- [Dynamic activity goals with Automatic Activity Detection](#)
- [Heart Health \(Cardiovascular Age, Cardio Capacity \(VO2 Max\)\)](#)
- [Metabolic Health \(Glucose Tracking, Meals\)](#)
- [Oura Labs](#)
- [Resilience](#)
- [Weekly, monthly, quarterly, yearly, and anniversary reports](#)
- [Women's Health \(Cycle Insights, Pregnancy Insights\)](#)

Body clock and chronotype, Explore content, and personalized activity goals are also available for Gen2 and Gen3 rings.

Regardless of the type of membership you have, your Oura Membership will continue to work seamlessly with Oura Ring 4. Since your membership is tied to your Oura account (not the ring itself), all of your data and benefits will carry over. Once you pair your new Oura Ring 4, you'll see all your data in the Oura App.

See [Oura Membership](#) for more information and pricing.

Oura Ring 4 vs. Oura Ring 4 Ceramic

	Oura Ring 4	Oura Ring 4 Ceramic
Design & Material	• Fully titanium exterior • Titanium interior • Recessed interior sensors at 0.3mm	• Ceramic exterior • Titanium interior • Recessed interior sensors at 0.3mm
	• Black	



Design & Material	• Fully titanium exterior • Titanium interior • Recessed interior sensors at 0.3mm	• Ceramic exterior • Titanium interior • Recessed interior sensors at 0.3mm
Finishes/Colors	• Black • Brushed Silver • Gold • Silver • Stealth • Rose Gold	• Cloud • Midnight • Petal • Tide
Sensing Platform	Accuracy is powered by Smart Sensing that dynamically selects the optimal signal path	Accuracy is powered by Smart Sensing that dynamically selects the optimal signal path
Sensors	• Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7	• Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7
Dimensions	• Width: 7.9mm • Thickness: ~2.88mm • Weight: 3.3-5.2 grams (depending on ring size)	• Width: 7.9mm • Thickness: 3.51mm • Weight: 5.1 - 8.1 grams (depending on ring size)

More Information

[Product Safety & Use](#)

[Oura Membership](#)

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[Oura Ring Finishes](#)



	- Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7	- Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7
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More Information

- Product Safety & Use
- Oura Membership
- Previous Oura Ring Generations
- Oura Ring Finishes
- Ring Battery Tips

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Product Safety & Use

This article outlines best practices, key information, and safety information to keep in mind while using your Oura product.

[General Use](#)[Memory Capacity](#)[Battery & Charging](#)[Care & Maintenance](#)[Ring Stuck?](#)[Safety Information](#)[Information on EMF](#)[Proposition 65](#)[Sustainability](#)[PSE Certification \(Japan\)](#)

General Use

You can wear your Oura Ring 24/7. It's built for all-day comfort.

We don't recommend wearing your Oura Ring adjacent to other rings as this may cause discomfort or unintended scratching.

We recommend wearing your Oura Ring on your non-dominant hand. By doing so, you'll limit friction (and scratches to your ring) with items and surfaces you grab and touch each day.

The Oura Ring is water-resistant up to 100 meters (~330 feet), meaning that you can wear it while showering, swimming, or partaking in recreational diving (e.g. *snorkeling*), but we don't recommend using your ring while scuba diving or keeping it submerged underwater for over 12 hours.

Oura Ring operating temperature ranges from -10–52°C / 14–125°F. You can safely wear your ring in the shower, hot tubs, saunas, ice baths, and cryotherapy tanks. However, extended exposure to extreme temperatures (below -20°C / -4°F or above 60°C / 140°F) may lead to battery damage.

For more product specifications, see [Oura Ring 4](#) and [Oura Ring Generation 3](#).

Data Security

Oura employs data encryption, which ensures secure pairing of your ring and app. This means your personal data will be kept fully secure in case an external party ever gets a hold of your ring.



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Oura App and Firmware

The Oura App is available on [the App Store](#) and [Google Play](#).

The Oura App is supported on iOS devices running Apple iOS 16 and higher, or Android devices running Android 10 and higher with Google Play services. Your device must support Bluetooth 4.0.

Ring [firmware updates](#) are accessible directly from your Oura App.

Memory Capacity

We recommend you sync your Oura Ring daily with your Oura App. However, the Oura Ring is capable of storing up to one week of data depending on the type and frequency of use. This means that your ring can continue to collect data without syncing to your device for up to one week. This would apply in the case that you wanted to enable [Airplane Mode](#) for an extended period of time.

Airplane Mode can only be enabled for a maximum of seven days. This is because Airplane Mode is automatically disabled each time your ring is placed back on its charger. Once Airplane Mode is disabled upon this action, a data sync between your ring and app will be initiated upon re-charging. If you'd like to keep Airplane Mode enabled for extended periods of time, you'll still need to charge your ring every few days for optimal battery performance and daily use. Be sure to re-enable Airplane Mode following the completion of charging—instructions on how to do so can be found in [this article](#).

Battery & Charging

Oura Ring has a rechargeable, non-replaceable 15mAh (US6) - 22mAh (US13) Lipo battery.

Charging in room temperature is recommended. Do not use or store the charger in humid environments like bathrooms.

See [Ring Battery Tips](#) and [How to Use Your Oura Charger](#) for more information.

Care & Maintenance

Cleaning Your Oura Ring

To clean your Oura Ring, you can rinse it with a small amount of mild dish soap and water. Remember to thoroughly dry your hands and your Oura Ring before placing it back on your finger.

Hand sanitizer is safe to use with the Oura Ring.



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Hand sanitizer is safe to use with the Oura Ring.

If you're following more stringent hand washing protocols, you can follow the same steps you would with other jewelry and remove your ring while scrubbing your hands to give the ring a separate cleanse.

Tips for Handling and Safety

- Don't leave the ring exposed to heat for extended periods
- Don't puncture the Oura Ring or its battery

Tips for Scratch Prevention

The Oura Ring may get scratched and can itself scratch softer metal jewelry or other objects in close contact with the ring, such as phone covers made of gold, silver, or aluminum. Some ceramic phone covers with a soft coating may also get scratched.

Scratches or other damage caused by wear and tear are not covered by Oura's [warranty policy](#). To help prevent scratches and increase the longevity of your ring:

- Wear your Oura Ring on your non-dominant hand to reduce the amount of contact with various surfaces and objects you interact with on a daily basis
- Avoid wearing other rings adjacent to your Oura Ring. Other rings may scratch your Oura Ring while rubbing against it
- Consider removing your Oura Ring in situations that involve friction (e.g. weightlifting), or when handling objects made of metal, ceramics, or stone (such as handling pots or pans)

The Rose Gold finish is specially designed to be a living color that will develop a darker patina over time. The exact patina effect depends [on the use and care of your ring](#).

Removing a Stuck Ring

Your finger size can vary depending on the time of the day, food and beverage you've recently consumed, rigorous exercise you've recently completed, and changes in altitude. In these cases, it may be difficult to remove the ring from your finger.

In case your ring gets stuck:

- Use cold water and gentle soap to wet your finger, and slowly twist the ring to remove it
- Hold your hand up above your heart until the finger swelling is reduced, and then try to remove it

In cases of emergency, discomfort, or if you can't remove the ring yourself, seek immediate medical attention



- Use cold water and gentle soap to wet your finger, and slowly twist the ring to remove it
- Hold your hand up above your heart until the finger swelling is reduced, and then try to remove it

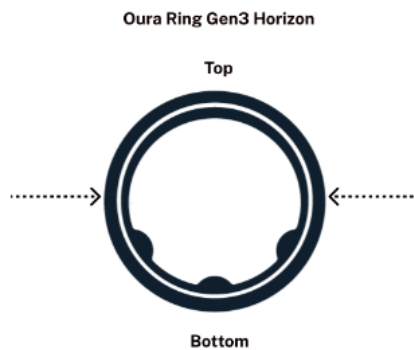
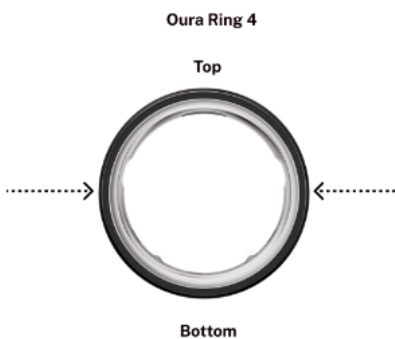
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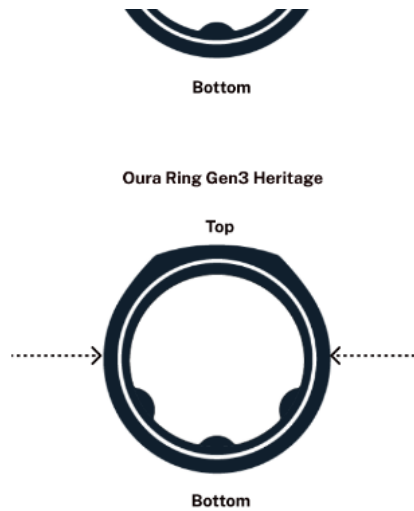
If a medical professional needs to cut the ring, here is some guidance to consider.

Note: These guidelines are for informational purposes and should only be executed by trained medical professionals with the appropriate tools. An option may be a tool like the Dolphin Cutter, an FDA-registered ring cutter.

Cutting a ring without the proper expertise can cause injury.

1. Orient the ring so the sensors are on the palm side of your finger. For Heritage, the flat plane should be on the top of your finger, and for Horizon and Oura Ring 4, the dimple should be on the palm side of your finger
2. Mark two cut locations on either side of the ring (at 3 & 9 o'clock) with a pen, marker, or tape
3. Rotate the ring to provide access to one of the cut locations
4. Use an appropriate medical-grade ring cutter capable of cutting titanium to make the first cut at the marked location
5. Rotate the ring if needed and cut the other location





Safety Information

General Safety

Oura products have been through various testing and analysis phases, and we continue to test them for safety.

The Oura Ring is not a medical device and is not intended to diagnose, treat, cure, monitor, or prevent medical conditions or illnesses. Please do not make any changes to your medication, nutrition, or workouts without first consulting your doctor or another medical professional.

Keep the ring away from children. This product is not intended for individuals under the age of 18. Seek immediate medical attention if you know or suspect that a child has swallowed the ring.

If you experience redness or skin irritation on your finger while wearing your Oura Ring, remove it immediately. If symptoms persist longer than 2-3 days, please contact a dermatologist.

Light Emission

The Oura Ring uses infrared (IR), red, and green light-emitting diodes (LEDs). These lights do not generate any appreciable heat or pose a health hazard.

Battery Safety

Avoid handling batteries, or working on devices with machinery that contain batteries, while wearing your Oura Ring. In certain cases, where both the cathode and the anode of another battery touch the ring, there is a risk of a short circuit which is similar to standard metallic rings. This can result in a potentially dangerous shock. Please take the proper precautions to avoid these situations.

Travel and Daily Use

Be cautious when moving heavy objects while wearing your ring to prevent it from getting caught on any fixed structures.

UV light won't affect the ring's materials



dangerous shock. Please take the proper precautions to avoid these situations.

Travel and Daily Use

Be cautious when moving heavy objects while wearing your ring to prevent it from getting caught on any fixed structures.

UV light won't affect the ring's materials.

The molding polymer used in the Oura Ring can handle short exposure to acetone.

You can wear the Oura Ring through airport X-ray machines, metal detectors, and scanners.

Information on EMF

The Oura Ring is a Bluetooth Smart®, class 2 device. Bluetooth is only active during a small portion of each day—well below 1%. Data is transmitted continuously when the ring is syncing with the app and during firmware updates.

The limit for Specific Absorption Rate (SAR) for a device like the Oura Ring is 2.0 W/kg for head and body. **The Oura Ring SAR level is 0.0003 W/kg.** For context, all cell phones sold in the United States tend to have a SAR level at or below 1.6 W/kg.

You can always turn on **Airplane Mode** if you'd like to eliminate all EMF exposure from your ring.

Proposition 65

Oura Ring 4 Ceramic:

- Based on current information, this product does not require a Proposition 65 warning.

Oura Ring Gen3 and Oura Ring 4:

- **WARNING:** This product may contain chemicals known to the State of California to cause birth defects or other reproductive harm.

For more information on Proposition 65 chemicals, please visit:

<https://www.p65warnings.ca.gov/>

Sustainability

We are being mindful of how we grow in a socially responsible manner. When it comes to what we currently have in place, we take pride in our focus on sustainability in the manufacturing of the Oura Ring. We have done a number of independent carbon footprint reports to measure our production emissions. Additionally, Oura only uses reputable suppliers and vendors who are compliant with local employee laws and social legislation. We have multiple ongoing initiatives within Oura and look forward to sharing more about them in the future.

If you're no longer using an old ring, we ask that you perform a factory reset on it



sustainability in the manufacturing of the Oura Ring. We have done a number of independent carbon footprint reports to measure our production emissions. Additionally, Oura only uses reputable suppliers and vendors who are compliant with local employee laws and social legislation. We have multiple ongoing initiatives within Oura and look forward to sharing more about them in the future.

If you're no longer using an old ring, we ask that you perform a factory reset on it first, and pass it along to a friend or family member if it's still in quality condition. Otherwise, we ask that you recycle it, or dispose of both the ring and internal battery in accordance with your local laws and regulations. Special handling may be required. The charger is an electronic device and local legislation for electronics recycling should be followed.

Read more about our [Device Recycling Program](#).

PSE Certification (Japan)

The Product Safety Electrical Appliance & Material Certification (PSE) certification is a mandatory product approval for electrical products in Japan. However, it only applies to products with removable lithium-ion batteries. Since the battery in the Oura Ring is built-in and not user-removable, this certification is not required.

Oura holds the MIC (Ministry of Internal Affairs and Communications) certification, which is Japan's required approval for products with radio technologies. Both the Oura Ring and its charger qualify as radio products under Japanese regulations.

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Blood Oxygen Sensing (SpO2)

The Oura Ring can detect overall blood oxygen saturation levels, and variation in those levels, while you sleep. These measurements show how efficiently your body circulates and absorbs oxygen, making it a useful index of your overall health, as well as indicating possible disturbances in your breathing during the night.

Blood Oxygen Sensing is available for:

- [Gen3](#) and [Oura Ring 4](#) with **active membership**
- [iOS](#)
- [Android](#)

This feature is **not** available on Gen2 or older.

[How Does Blood Oxygen Sensing Work?](#)

[How to Use the Blood Oxygen Sensing Features](#)

[Blood Oxygen Sensing Troubleshooting](#)

[Things to Keep in Mind](#)

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How Does Blood Oxygen Sensing (SpO2) Work?

The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDs into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light.

Your blood oxygen measurements are located on the Oura App's Sleep screen. The **Average Blood Oxygen** reading is the percentage of your blood that contains oxygen. The **Breathing Regularity** graph displays variations in how much oxygen is present in your blood. This is different from [respiratory rate](#), which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during **sleep periods longer than three hours**, but the Oura App only displays the results for the longest sleep period of the day.

Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day.

How to Use the Blood Oxygen Sensing (SpO2) Features



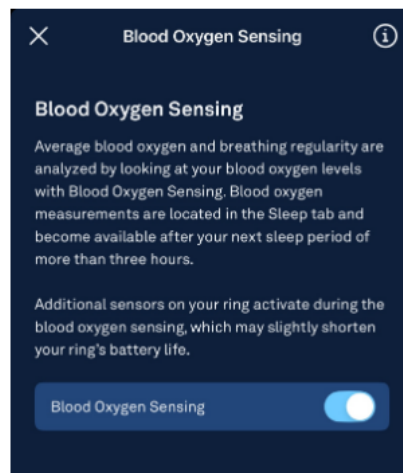
the day.

Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day.

How to Use the Blood Oxygen Sensing (SpO2) Features

Average Blood Oxygen and Breathing Regularity are turned on by default. To toggle them on or off:

1. Go to the  **menu** in the upper left corner of the Oura App's home screen
2. Select **Blood Oxygen Sensing**
3. Select the toggle next to **Blood Oxygen Sensing**



Note: Your ring must be connected to the app over Bluetooth in order to switch the feature on and off.

After your next sleep period of more than three hours, check the Sleep screen. There you will find the **Average Blood Oxygen** percentage and **Breathing Regularity** display.



Suspected variations in your blood oxygen levels are shown in a color-coded timeline, indicating sampling intervals of 15 minutes.

- Optimal: no significant variations to display
- Good: few variations detected (dark blue line)



Suspected variations in your blood oxygen levels are shown in a color-coded timeline, indicating sampling intervals of 15 minutes.

- Optimal: no significant variations to display
- Good: few variations detected (dark blue line)
- Fair: occasional variations detected (light blue line)
- Pay attention: frequent variations detected (white line)

If your blood oxygen levels are measured as "Optimal" throughout the night, the Oura App will not display a Breathing Regularity timeline. Tap on the information ⓘ icon in the app to learn more about Breathing Regularity, how to improve it, and how to interpret the displayed information.

Blood Oxygen Sensing (SpO2) Troubleshooting

You might see **Average Blood Oxygen** values that are lower than expected for a variety of reasons, including:

- Sleeping in a position that constrains the amount of blood flow to your hand or arm
- Having the ring in the wrong orientation
- Having a too loose or tightly fitting ring
- Excessive movement during sleep

If you are not seeing any blood oxygenation data after a sleep period of three hours or more, try the following:

- Ensure that you have the latest **app software** and **ring firmware** installed
- Charge your ring to at least 30% before going to bed
- Ensure that your ring fits snugly, and that the LED sensors are on the underside (palm-side) of your finger
- Make sure that your ring is not in **Airplane Mode**
- Turn on your device's Bluetooth to enable your ring to sync data with the Oura App

Things to Keep in Mind

High variation in overnight blood oxygen measurements can be a sign that something is challenging your health, such as illness, altitude, or breathing issues.

The Oura Ring is not a medical device and is not intended to diagnose, treat, cure, monitor, or prevent medical conditions or illnesses. Please do not make any changes to your medication, nutrition, or workouts without first consulting your doctor or another medical professional.

The Blood Oxygen Sensing features use the red and infrared LEDs throughout the night, which impacts your ring's overall battery life. We recommend **turning on low battery notifications** as a reminder.

Blood Oxygen Sensing features do not directly impact any your daily scores.



The Oura Ring is not a medical device and is not intended to diagnose, treat, cure, monitor, or prevent medical conditions or illnesses. Please do not make any changes to your medication, nutrition, or workouts without first consulting your doctor or another medical professional.

The Blood Oxygen Sensing features use the red and infrared LEDs throughout the night, which impacts your ring's overall battery life. We recommend [turning on low battery notifications](#) as a reminder.

Blood Oxygen Sensing features do not directly impact any your daily scores. However, if something is impacting the variations in Breathing Regularity and Average Blood Oxygen, it may show up in the Sleep and Readiness scores already.

More Information

[Manage Notifications](#)

[An Introduction to Respiratory Rate](#)

[Blood Oxygen Sensing \(SpO2\) on Oura](#)

[10 Ways to Increase Your Blood Oxygen \(SpO2\) Levels](#)

[How We Developed the Blood Oxygen Sensing \(SpO2\) Feature](#)

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Live Heart Rate

Live Heart Rate lets you track your heart rate in real-time. This feature is available for:

- [Gen3](#) and [Oura Ring 4](#) with **active membership**
- [iOS](#)
- [Android](#)

This feature is **not** available on Gen2 or older.

[What Is Live Heart Rate?](#)

[How to Use Live Heart Rate](#)

[Things to Keep in Mind](#)

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What Is Live Heart Rate?

With Live Heart Rate, you can check your heart rate at any time during the day. Just tap to see your heart rate instantly. You can access Live Heart Rate through the 24-hour [heart rate graph](#).

Oura uses green LEDs to measure your heart rate in real-time. The process uses photoplethysmography (PPG) technology at a frequency of 50Hz.

PPG technology works by shining a light onto the surface of your skin. When this reaches your blood vessels, some of the light is reflected back onto a photodetector. The intensity of light reflected back captures the amount of blood in your blood vessel via a waveform.

Depending on the color of light used, different biometrics can be captured through their unique waveforms. For example, when your blood vessels are expanded, more green light will be absorbed compared to when they're contracted. This phenomenon allows your pulse to be accurately taken.

To gather an accurate pulse reading, you must remain relatively still for Live Heart Rate to take its full 10-second measurement. If Oura detects too much motion, your Live Heart Rate won't be available.

How to Use Live Heart Rate

1. Go to the Today screen to view your 24-hour heart rate graph. You'll see your most recent heart rate reading along with how recently it was updated.



How to Use Live Heart Rate

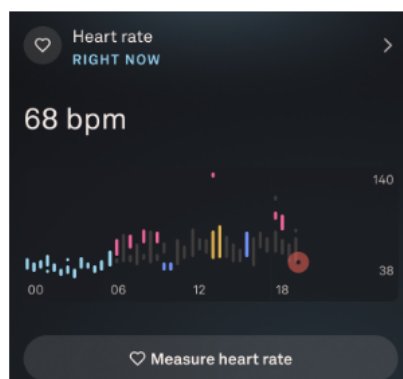
1. Go to the Today screen to view your 24-hour heart rate graph. You'll see your most recent heart rate reading along with how recently it was updated.



2. Tap the large heart icon. If the conditions are right (balanced skin temperature, low motion, and a strong Bluetooth connection), it'll begin loading. Stay still for 10 full seconds or until your measurement is complete.



3. Once the reading is complete, your heart rate graph will update with your current heart rate in beats per minute (BPM).



Things to Keep in Mind

- Several variables can interfere with a Live Heart Rate reading:
 - Poor signal quality: check your device's Bluetooth settings to confirm your ring and app are properly connected
 - Low skin temperature: unusually low skin temperature may interfere with



- Poor signal quality: check your device's Bluetooth settings to confirm your ring and app are properly connected
- Low skin temperature: unusually low skin temperature may interfere with readings
- Excessive movement: make sure you hold still for 10 seconds during your measurement
- Incorrect ring orientation: make sure your ring's sensors always remain in contact with your skin, on the palm side of your finger
- Live Heart Rate is not the same as [Activity Heart Rate](#), nor is it intended to be a constant heart rate monitor
- The green LEDs used to take your Live Heart Rate are designed for accuracy but require a stronger signal quality. Frequent use of Live Heart Rate will consume more battery, so we recommend charging your ring regularly to keep the battery in its optimal range. You can learn more about [charging best practices here](#).

More Information

[Heart Rate Graph](#)

[Activity Heart Rate](#)

[Ring Battery Tips](#)

[Oura Launches Daytime Heart Rate](#)

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Resting Heart Rate

Learn more about the Resting Heart Rate (RHR) measurements taken by the Oura Ring.

[What Is Resting Heart Rate?](#)
[How Oura Measures Resting Heart Rate](#)
[When Does Oura Measure Resting Heart Rate?](#)
[Interpret Your Resting Heart Rate](#)

What Is Resting Heart Rate?

Resting Heart Rate (RHR) is the number of times your heart beats per minute while you're at rest. It provides a snapshot of your sleep quality, recovery, stress response, activity level, and overall health.

Normal RHR in adults can range anywhere from 40-100 beats per minute (BPM). Oura develops an individualized baseline for your RHR so you can quickly notice when something is off.

Variation beyond 3-5 BPM above or below your personal average may be a sign of low recovery or excessive stress on your body, such as illness, poor diet choices, late-night eating, or lack of sleep.

The Importance of Resting Heart Rate

Resting heart rate is an indicator of both long and short-term health.

- **In the long term**, a low RHR is generally a sign of better cardiovascular health, physical fitness, and recovery
- **In the short term**, a high RHR can indicate behaviors you can control, such as internal stress, eating patterns, or exercise habits. These can be positive or negative, depending on the context. For example, completing a difficult workout is a positive precursor to an elevated RHR. Meanwhile, undergoing mental distress from work is generally viewed as a negative source of rises in RHR

A Lower RHR is a Good Sign

With each heartbeat, your body uses the oxygen-rich blood collected from your lungs to support functions throughout your systems. When your resting heart rate lowers over time, each heartbeat becomes more effective. This allows your body to accomplish the same amount of work necessary to keep itself running with less effort.



With each heartbeat, your body uses the oxygen-rich blood collected from your lungs to support functions throughout your systems. When your resting heart rate lowers over time, each heartbeat becomes more effective. This allows your body to accomplish the same amount of work necessary to keep itself running with less effort.

As your heart becomes more efficient, you're able to do more without increasing your heart rate, allowing your body to take on less strain.

With less strain, your body moves closer to a parasympathetic ("rest-and-digest") state. The more that your parasympathetic nervous system dominates, the more room there is for your sympathetic nervous system (fight-or-flight) to come into action when needed. This might be necessary when you decide to exercise, give a presentation, interview for a new job, try something for the first time, or have to meet a deadline at work.

How Oura Measures Resting Heart Rate

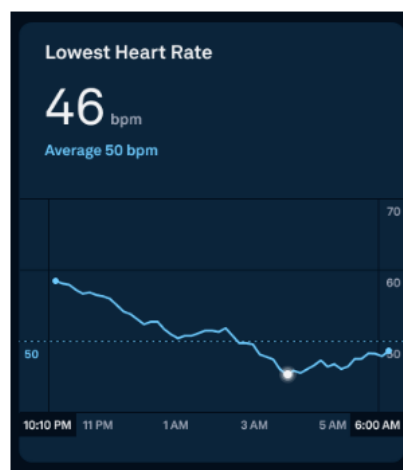
Oura monitors your RHR by detecting changes in your blood pulse volume through infrared photoplethysmography (PPG) sensors found inside your ring.

Each time that your heart beats, blood is pumped out to the arteries located in your hands and fingers. The PPG sensors are able to detect these changes in blood flow and volume using light reflection. Each pulse causes the arteries in your finger to alternate between swelling and contracting. By shining a light on your skin, changes in light reflected back from the wavering volume of red blood cells in your arteries are accounted for. From here, PPG can represent these blood flow changes through a visual waveform that represents the activity of your heart.

Oura provides you with the following RHR metrics:

- **Average RHR:** Your average RHR captured during the night
- **Lowest RHR:** Your lowest RHR value captured during the night

Your RHR values are captured every 10 minutes throughout the night. You can find your RHR in your Readiness and Sleep detail views and in **Trends**.





When Does Oura Measure Resting Heart Rate?

The Oura Ring Gen2 doesn't track heart rate during exercise or other activities during the day. The only exception is during **unguided sessions**, which allow you to meditate or check in with your body. Certain unguided sessions, upon completion, show you a snapshot of your RHR during the day.

Also, read more information about [our 24-hour heart rate graph](#).

Interpret Your Resting Heart Rate

Low vs. High, Compared to Your Average

If your lowest RHR of the night is roughly 0-10 BPM below your personal long-term average (collected over the past ~ 2 months), this is evidence that your body has recovered well and is in an optimal state to perform that day.

However, an exceptionally low resting heart rate may indicate that you are in a state of low nervous system arousal or lethargy after a very strenuous physical peak performance.

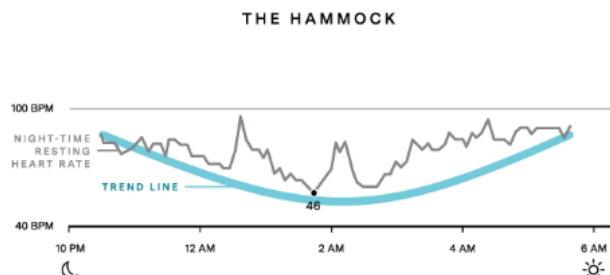
You'll observe decreased Readiness Scores if you receive an RHR that is 10-15 BPM lower than usual, as well as if you receive an RHR that is 3-5 BPM higher than usual.

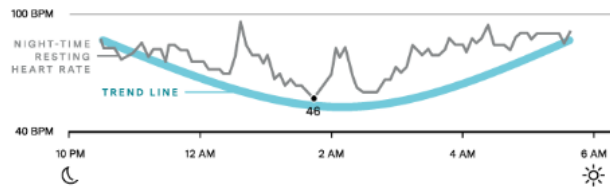
An elevated RHR is a sign that something is challenging your system to a degree that's leading to poor recovery. Both of these scenarios are signs to pay attention and give your body the care and time that it needs to recover properly.

Patterns in Your RHR

If your RHR reaches its lowest point during the first half of your night, it can be a sign you've recovered well from the previous day.

This ideal pattern is known as "The Hammock," which you'll see demonstrated below. For information about other RHR patterns that can occur at night, see [Sleeping Heart Rate: Look for These 4 Patterns](#).





This ties into the concept of heart rate stabilization, which divides sleep into two phases: repair and recharge.

During repair, your body recovers from the previous day. During recharge, your body prepares for the coming day. Once your heart rate reaches its lowest point during the night (known as stabilization), your body transitions from repair to recharge. The sooner your heart rate can reach its true nighttime low, the more time your body has to refuel and get ready for the day ahead.

More Information

[Your Resting Heart Rate](#)

[How Accurate Are Oura's Heart Rate & HRV Measurements?](#)

[Sleeping Heart Rate: Look for These 4 Patterns](#)

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Heart Rate Variability

Learn more about the heart rate variability (HRV) measurements taken by the Oura Ring.

[What Is Heart Rate Variability?](#)[How Oura Calculates HRV](#)[The Importance of HRV](#)[When Does Oura Measure HRV?](#)[More Information](#)

What Is Heart Rate Variability?

Heart rate variability (HRV) reveals how your body responds to training, traveling, or stress. It's a measure of the millisecond-level variation in time between your heartbeats and offers insight into the balance between the two divisions of your autonomic nervous system:

- Sympathetic: the "fight-or-flight" response
- Parasympathetic: the "rest-and-digest" mechanism

HRV is a personalized metric, with typical values ranging from under 20 to over 200 milliseconds. HRV is influenced by various factors like age, health status, and fitness level. It's important to compare your current HRV only against your own historical trends, not against others.

Interpreting your HRV

When compared to your individual baseline:

- **High HRV** is associated with activation of your "rest-and-digest" system, general physical fitness, and good recovery. Higher average nighttime HRV is also associated with better sleep quality
- **Low HRV** is associated with activation of your "fight-or-flight" system. It's commonly linked to stress (both good and bad), illness, and overtraining.

How Oura Calculates HRV

Oura provides a reliable and accurate measurement of your autonomic nervous system's activity by calculating your nighttime HRV.

Oura takes five-minute samples, not just a single reading, throughout the night. This continuous tracking offers greater accuracy and a more detailed view of your body's response to stress and preparation for the day ahead. The average HRV

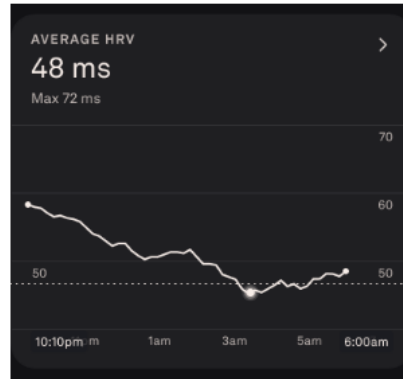


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Oura takes five-minute samples, not just a single reading, throughout the night. This continuous tracking offers greater accuracy and a more detailed view of your body's response to stress and preparation for the day ahead. The average HRV value you see is the mean of all five-minute samples taken while you sleep.

Key HRV metrics:

- **Average HRV:** Your average HRV measured over the entire night
- **Max HRV:** Your highest HRV value recorded during the night



You can monitor your HRV under the [Readiness](#) or [Trends](#) views. Tracking HRV in the Trends view is particularly helpful for assessing your recovery over the past week and identifying any internal or external stressors that may require attention.

The Importance of HRV

HRV reflects how the sympathetic and parasympathetic branches of your autonomic nervous system work together.

When both of these branches send signals to your heart, your heart rate naturally increases and decreases, resulting in variations between beats. Heart rate and HRV tend to be inversely connected. When your heart rate is slower, there's more time for variation between beats, leading to a higher HRV. On the other hand, a faster heart rate leaves less room for variation, usually resulting in a lower HRV.

Having variation between your heartbeats is a sign that your autonomic nervous system is balanced and capable of responding to different stimuli in a healthy way.

When Does Oura Measure HRV?

Oura measures HRV only during sleep to ensure the most reliable and interpretable data. Heart rate and HRV are highly sensitive metrics that can be easily affected by simple daytime activities—such as drinking water, getting up to use the bathroom, or watching an exciting TV show—which would introduce "noise" and make the data difficult to analyze.

Gen2 users: Certain [unguided sessions](#) offer snapshots of heart rate, HRV, and



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Gen2 users: Certain [unguided sessions](#) offer snapshots of heart rate, HRV, and skin temperature upon completion, and can be used any time of day.

Gen3 and Oura Ring 4 members: [Explore](#) contains meditation sessions and stories, some of which will also measure your heart rate, HRV, and skin temperature.

More Information

[What Is Heart Rate Variability?](#)

[How to Increase Your HRV \(Heart Rate Variability\)](#)

[How Accurate Are Oura's Heart Rate & HRV Measurements?](#)

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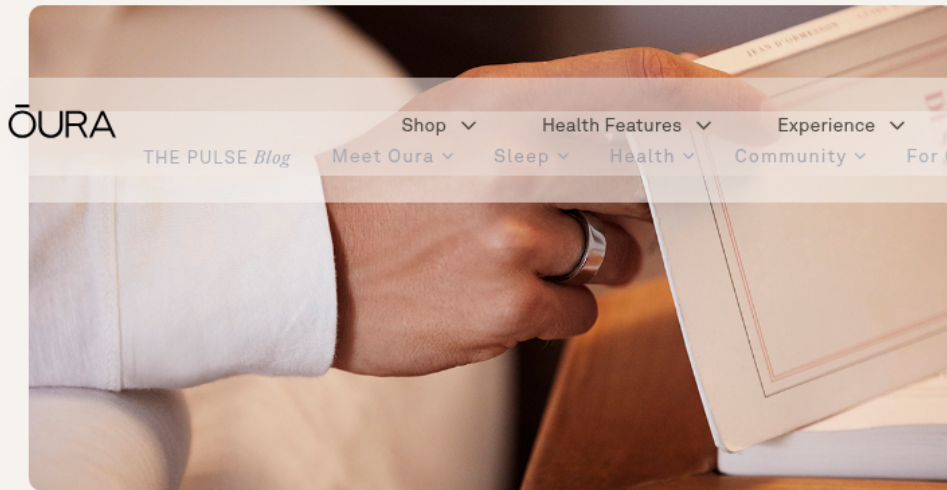
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Heart Health

How Oura Measures Respiratory Rate From Your Finger



Author: Oura Team | February 6, 2024



Respiratory rate can be measured directly by counting the rise and fall of someone's chest or indirectly by reading it from another signal in your body, like your heart's activity. A widely used method is deriving respiration from photoplethysmography (PPG), the same light pulses used to measure heart rate in Oura and many hospital devices.

How Does It Work?

Oura derives your respiratory rate exclusively from your **nighttime PPG data** – the same signal it uses to measure your **resting heart rate** and **heart rate variability**.

Nighttime is an ideal time to measure respiration because your body is in a consistent state. During the day, many factors can influence your respiratory rate (e.g., **caffeine**, **emotion**, **exercise**), and it's even been shown that your rate increases when you know it is being counted.

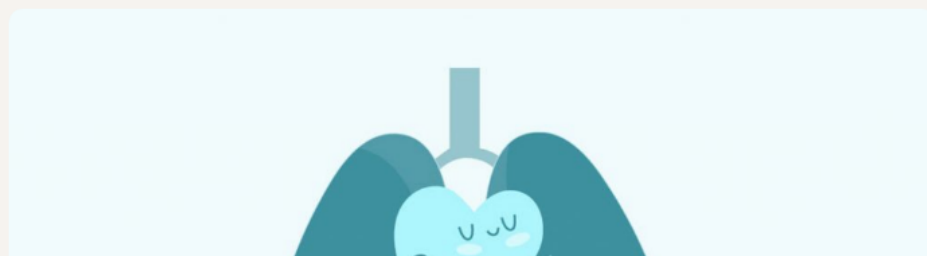
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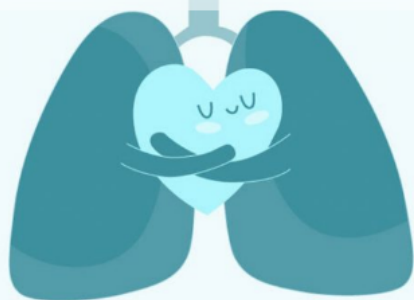
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You can measure respiration in your heartbeat because your heart and lungs are closely connected by a process called respiratory sinus arrhythmia. Although it sounds serious, it's a completely normal and simple partnership:

- When you breathe in, your heart rate increases.
- When you breathe out, your heart rate slows down.

This helps your body take in oxygen and expel CO2 with as few heartbeats as possible. Because this pattern is predictable, you can measure the variation between heartbeats to determine your respiratory rate.

READ MORE: [Understanding Your Respiratory Rate While Sleeping](#)

How Is It Calculated?

Your Oura uses your PPG data to spot your respiratory sinus arrhythmia pattern (your heart speeds up as you breathe in and slows down as you breathe out). Your respiratory rate is calculated based on the number of times this happens in a minute.

Let's explore how that looks in the data.

1. Electrical signals power your heart to contract. This is what you see in a classic ECG signal showing a few heartbeats:

ECG



2. Each time your heart beats, it sends a pulse of blood moving out through your arteries, including the ones in your finger where your Oura sits. You can read this activity without an ECG by using a PPG sensor to shine a light onto your skin. As

2. Each time your heart beats, it sends a pulse of blood moving out through your arteries, including the ones in your finger where your Oura sits. You can read this activity without an ECG by using a PPG sensor to shine a light onto your skin. As the pulse of movement through your circulatory system changes, the amount of light that is reflected back changes with it, signaling precisely how your heart is beating. Oura uses infrared PPG which penetrates deeper than other green light wearables to measure as close to your arteries as possible.

PPG



3. In this PPG data, your respiratory sinus arrhythmia pattern can be spotted by identifying when your heart rate speeds up (breathing in) or slows down (breathing out). This is done by calculating the interval between your heart beats, your interbeat intervals (IBI). You can follow that process in-depth with the image below:

PPG Signal Is Collected

PPG Signal Is Used to Calculate IBI



IBI Is Mapped out Over Time



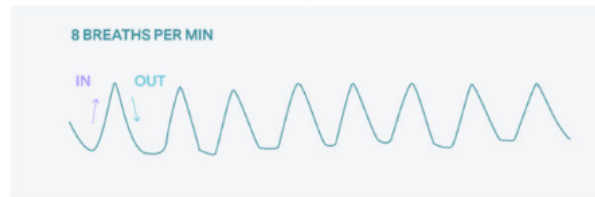
Decreasing IBI Periods Are ID-ed as Breathing In, Increasing IBI Periods as Breathing Out



Final Respiratory Rate Is Calculated



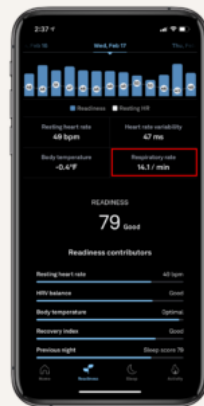
Final Respiratory Rate
Is Calculated



READ MORE: [What is the Relationship Between Heart Rate and Respiratory Rate?](#)

How To Interpret Your Respiratory Rate

Currently, the Oura App gives the [average respiratory rate](#) from your night under the [Readiness](#) tab.



By tapping on this number, you can also expand this view to look at your trend over time. This view answers the question, "Is my breathing rate increased compared to my baseline?" or more widely, "Is something challenging my system?"

A typical range for a healthy adult is 12–20 breaths per minute, and a fluctuation of more than 1-2 breaths per minute is notable.



Keep in mind: PPG respiratory rate is a great way to compare your averages across different nights.

For most people, respiratory rate can be easily determined over the course of the





Keep in mind: PPG respiratory rate is a great way to compare your averages across different nights.

For most people, respiratory rate can be easily determined over the course of the night. However, there are some breathing conditions that may disrupt portions of the nightly signal. In most cases, this will not impact the accuracy of their data, but is worth noting.

For example, the heart of an individual with sleep apnea may keep beating in a pattern that anticipates a normal respiratory sinus arrhythmia (breathe in – speed up, breathe out – slow down). But the actual air flow can be restricted for the individual and may result in a different number of breathing cycles than can be seen in the PPG signal. For most individuals, this is not a problem, especially when you look at [long-term trends](#) and deviations from your baseline.

RELATED: [How Accurate Is Oura's Respiratory Rate?](#)

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English



Cypress PSoC® 6 Microcontrollers

Purpose-Built for the Internet of Things

WWW.CYPRESS.COM/PSOC6



INCREASING DEMANDS FOR THE INTERNET OF THINGS

The IoT is exploding, with more than 30 billion devices projected to be in service by 2020. At the heart of every IoT device lies a mixed-signal embedded system that is taking on an increased amount of data processing, fueled by cloud connectivity. Devices such as door locks, factory machinery, and wearables are becoming smart — sensing, connecting, learning and responding — to make life easier. With connectivity, security becomes critical to protect users from malicious activity. These next-generation IoT devices require increased processing and security without incurring power and cost penalties.

Increased Data Processing

Today's IoT devices are collecting and processing more data than ever. Even wearable devices can contain more than a dozen sensors, including biometric, environmental, and motion sensors. Sensor data is collected and processed in real-time requiring a processor capable of sensor aggregation.

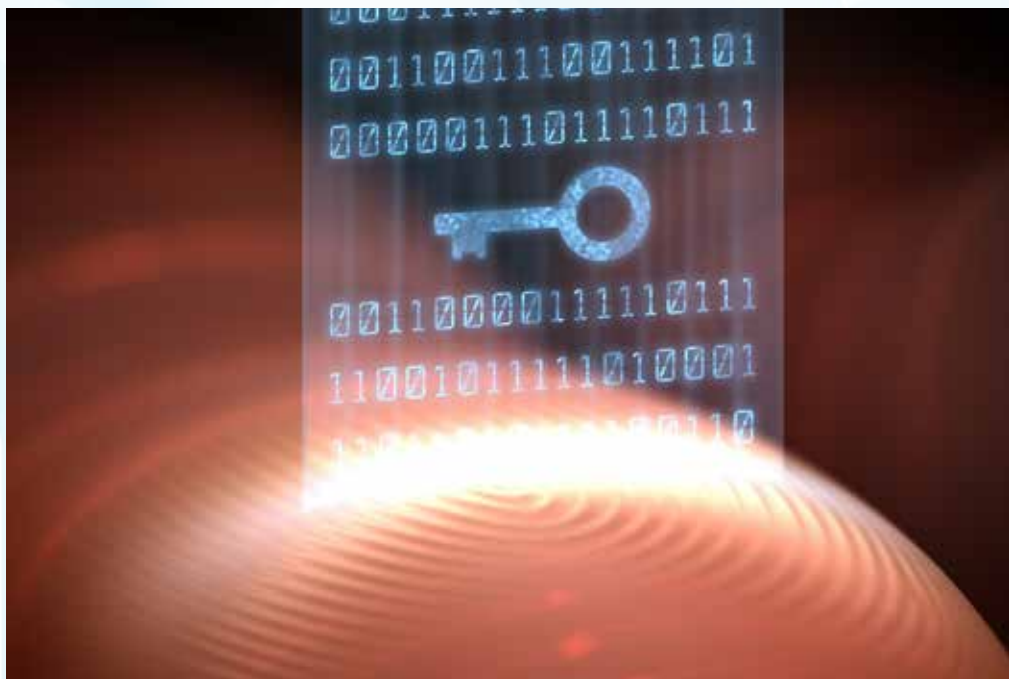


Flexible Connectivity

IoT devices connect to the cloud via various wired and wireless protocols including USB, Bluetooth Low Energy (BLE), Wi-Fi, and LPWAN. Having the flexibility to connect via different connectivity options is essential.

Security

Security and privacy are significant IoT concerns. Users must be confident that their personal data is always protected while their IoT devices are connected to the Internet. Similarly, IoT developers must be able to ensure the authenticity of their products to eliminate counterfeits and perform secure over-the-air updates to keep their products current.



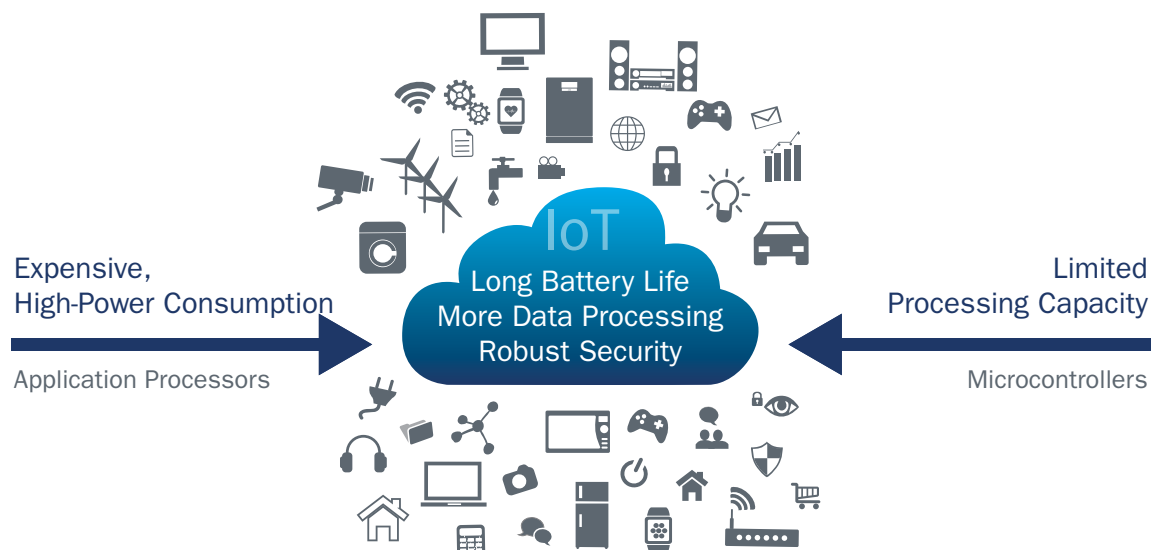
Extended Battery Life

From wearables to remote sensors, more and more IoT devices are running on batteries. Extending the time between battery charges is a key requirement for battery-powered IoT devices.

PSoC® 6 MICROCONTROLLERS

Purpose-Built for the Internet of Things

Cypress' PSoC 6 MCU architecture is purpose-built for the IoT, filling the gap between expensive, power-hungry applications processors and low-performance MCUs. The ultra-low power PSoC 6 MCU architecture offers the processing performance needed by IoT devices. Security is built-in, enabling a single-chip solution. PSoC 6 enables engineers to uniquely create innovative, next-generation IoT devices leveraging the unique PSoC fabric with its easy-to-use, software-defined peripherals.



LOWEST POWER

IoT devices are often portable, making battery life a critical factor. The PSoC 6 MCU architecture is built on a cutting-edge, ultra-low-power, 40-nm process technology with a dual Arm® Cortex®-M core architecture. Active power consumption is as low as 22-µA/MHz for the M4 core, and 15-µA/MHz for the M0+ core. PSoC 6 delivers extended battery life without sacrificing performance.



MOST FLEXIBILITY

The rapid growth of the IoT is sparking a need for innovation in IoT products. The PSoC 6 MCU architecture's best-in-class flexibility enables the addition of new features, and addresses the need for unique IoT products with multiple connectivity options, such as USB and BLE. PSoC 6 also offers software-defined peripherals to create custom analog and digital circuits, and the industry's best capacitive-sensing solution, CapSense®. In addition, a flexible dual-core architecture is used to optimize for system power consumption and performance. The possibilities are endless.



BUILT-IN SECURITY

With a growing number of devices connecting to the IoT, security must be established between hardware, cloud applications, and servers, and finally users and services. The PSoC 6 MCU architecture supports multiple, simultaneous secure environments without the need for external memories or secure elements, and offers scalable secure memory for multiple, independent user-defined security policies, preventing your IoT device from becoming a security liability. PSoC 6 provides you with a new standard for IoT security.

UNMATCHED SOLUTIONS FOR THE INTERNET OF THINGS

The Oura Ring

OURA

The Oura Ring is the world's most-advanced wearable that helps you achieve restorative sleep. This innovation is delivered through PSoC 6 MCUs, enabling the following features:

SMALL FORM FACTOR

Integration of several functions into a single PSoC 6 MCU helped reduce the size of the ring over the prior generation by 50%.

DATA STORAGE

PSoC 6's on-chip flash memory enables storage of up to six weeks of user data.

BLUETOOTH LOW ENERGY

The built-in BLE wireless connectivity in PSoC 6 provides communication to the mobile app device.



EXTENDED BATTERY LIFE

PSoC 6's ultra-low power was able to extend battery life to seven days, 3X longer than the prior generation ring.

SENSOR FUSION

PSoC 6, as the host processor, connects different biometric sensors in the ring to collect and process data.



DATA PROCESSING

The dual Cortex-M4 and Cortex-M0+ cores in PSoC 6 enabled 10X more processing capability vs. the prior-generation ring.

"The Oura ring required long-lasting battery life, increased processing capacity, and the use of advanced sensors, all in a stylish and comfortable form factor. Choosing Cypress as our partner helped us meet these requirements and redefine what is possible for wearable health products."

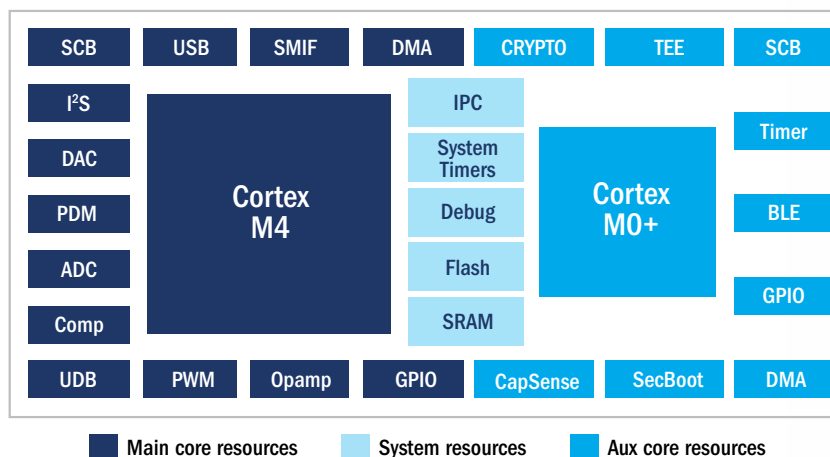
Petteri Lahtela
CEO and Co-founder at Oura Health

For more information on the Oura Ring, please visit: www.ouraring.com

PSoC 6 MCU PORTFOLIO

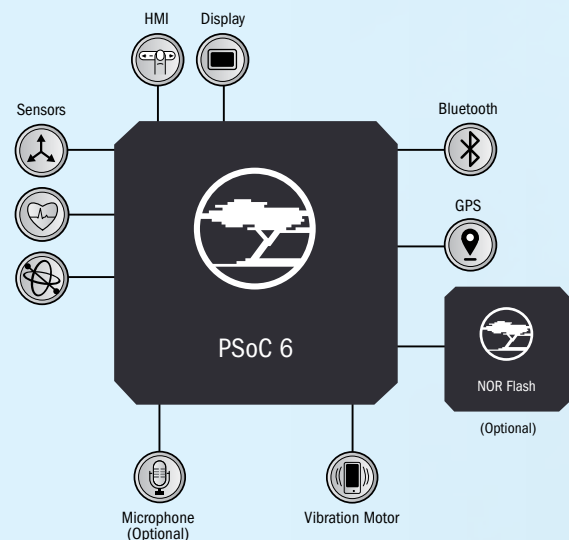
Value Line	Programmable Line	Performance Line	Connectivity Line	Security Line
PSoC 60	PSoC 61	PSoC 62	PSoC 63	PSoC 64
Arm Cortex-M4 (50 MHz)	Arm Cortex-M4 (150 MHz)	Arm Cortex-M4 (150 MHz) & Cortex-M0+ (100 MHz)	Arm Cortex-M4 (150 MHz) & Cortex-M0+ (100 MHz)	Arm Cortex-M4 (150 MHz) & Cortex-M0+ (100 MHz)
Up to 2MB Flash / 1MB SRAM	Up to 2MB Flash / 1MB SRAM	Up to 2MB Flash / 1MB SRAM	Up to 1MB Flash / 288KB SRAM	Up to 2MB Flash / 1MB SRAM
				Rich Execution Environment Secure Execution Environment Hardware-based Root-of-Trust and Trusted Services
				BLE
				ULP/LP
				CapSense, UDBs, Opamps, Crypto, PDM-PCM
				SCBs, TCPWMs, QSPI, I2S, USB, DMA, SAR ADC, DAC, LCD, LP COMP, RTC, Efuse, PLLs, OSCs, GPIOs, SmartIO

PSoC 6 DUAL-CORE MCU ARCHITECTURE



WEARABLE EXAMPLE

Modern wearables incorporate a growing number of functions and capabilities while maximizing battery life. Functions such as sleek user interfaces with sliders and proximity sensors, voice command support, seamless BLE connectivity, display driving, and process-intensive sensor aggregation can all be achieved in a single PSoC 6 MCU without sacrificing battery life.



KEY FEATURES



LOW POWER

PSoC 6 was designed for ultra-low power consumption. Starting with the dual-core architecture, the Cortex-M4 core can be put to sleep to conserve power, while monitoring functions can be managed by the Cortex-M0+ core. Dynamic voltage and frequency scaling can be controlled by the user to optimize power for different operating environments. With multiple sleep and hibernate modes, users can fine-tune their power profiles.



BUILT-IN SECURITY

PSoC 6 is designed to provide a hardware-based root of trust for secure applications. A hardware-based root of trust provides extra protection against unauthorized access to keys. In addition, hardware accelerated cryptography and true random-number generation are provided, which are needed for secure boot, authentication, secure over-the-air updates and other functions.



CONNECTIVITY

PSoC 6 supports multiple wired and wireless connectivity options including Bluetooth Low Energy, Wi-Fi, and USB.



CAPSENSE

PSoC 6 leverages Cypress' industry-leading CapSense solution for capacitive sensing with state-of-the-art noise immunity and water rejection used for applications including touch buttons and sliders, proximity detection and liquid-level sensing.



SOFTWARE-DEFINED PERIPHERALS

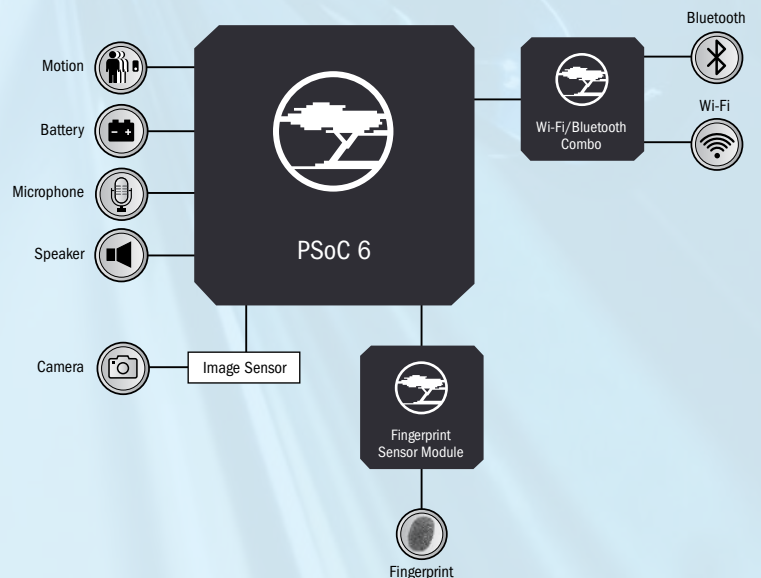
Cypress' software-defined peripherals can be configured using pre-built library functions or customized using the programmable architecture inside of PSoC 6 devices:

Programmable analog blocks are composed of an assortment of opamps, comparators, ADCs, and DACs, enabling complex analog signal flows.

Programmable digital blocks are composed of Universal Digital Blocks (UDBs), Serial Communication Blocks (SCBs), and Timer / Counter / Pulse Width Modulators (TCPWMs). These blocks can be configured to set-up custom digital interfaces, state machines, and custom logic functions.

VIDEO DOORBELL EXAMPLE

For a smart home application like a video doorbell, PSoC 6, with its dual Cortex-M4 and Cortex-M0+ cores, can transmit real-time video as well as incorporate proximity detection, capture and play audio, and interface to a Cypress fingerprint sensor module or Cypress WICED Wi-Fi chipset. The PSoC 6 MCU's ultra-low-power profile enables smart applications to run off batteries.



ACCELERATE YOUR NEXT IOT DESIGN CYCLE

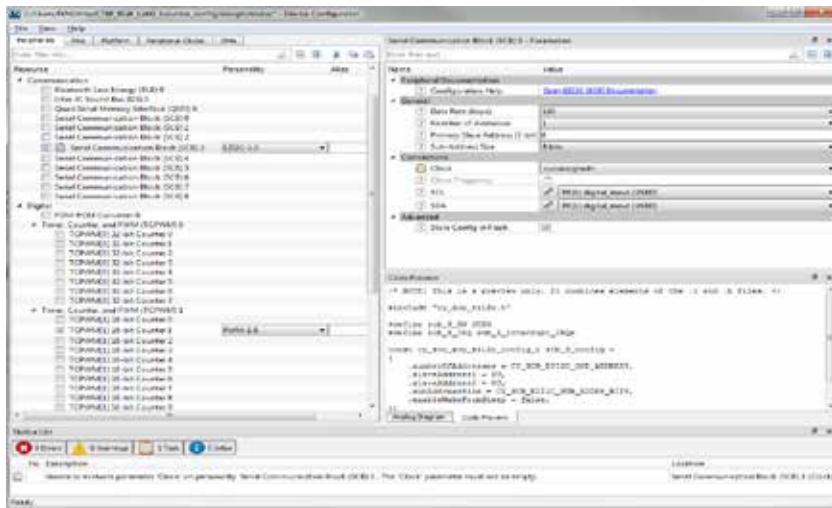
ModusToolbox™ Software Environment

ModusToolbox simplifies development for IoT designers, delivering easy-to-use tools and a familiar IDE, with support for Windows®, macOS®, and Linux. It provides a powerful suite of tools for system and peripheral set-up. Code examples, documentation, RTOS and Cloud service support, and technical support forums are available to help your IoT development process along.

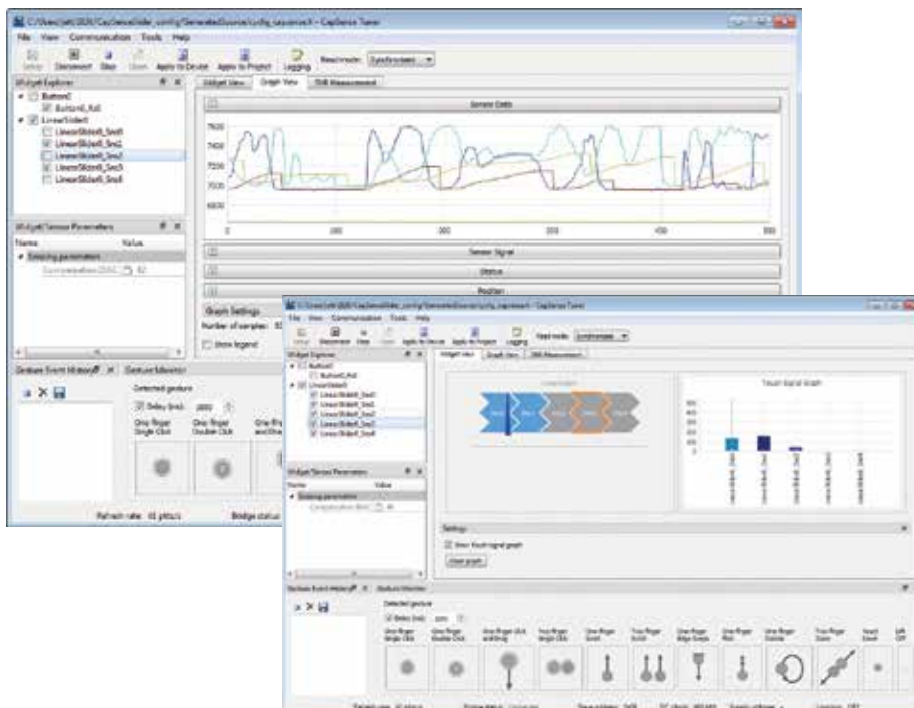


Graphical Configuration Tools:

GUI-based configuration tools used to configure peripherals such as Smart I/O, CapSense, and wireless connectivity to quickly design and develop IoT systems.



PSoC 6 MCU Peripherals Configurator



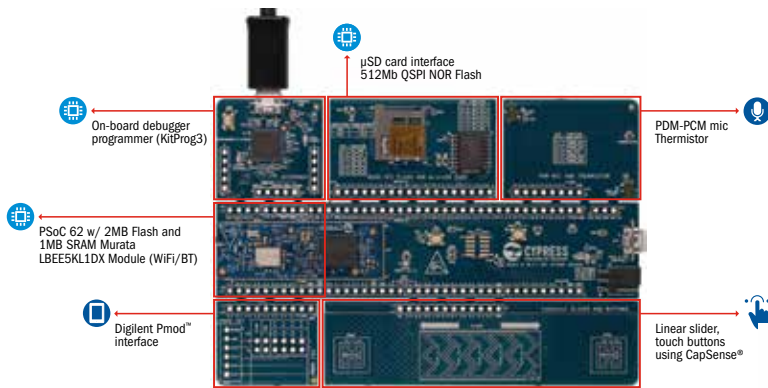
CapSense Configurator Tool

To start designing with the ModusToolbox Software Environment, please visit www.cypress.com/ModusToolbox

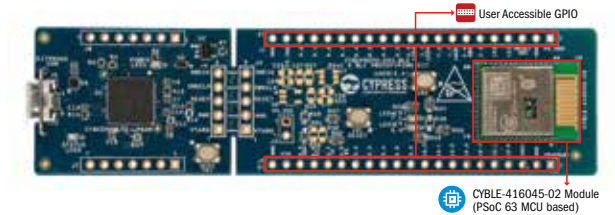
PSoC 6 DEVELOPMENT KITS

Cypress offers award winning development kits along with hundreds of code examples allowing you to evaluate, prototype and debug designs with PSoC 6. The PSoC 6 development kits are pin-compatible with various 3rd party expansion boards including Arduino™ shield boards and Digilent® Pmod™ daughter cards.

PSOC 6 WI-FI BT PROTOTYPING KIT
(CY8CPROTO-062-4343W)



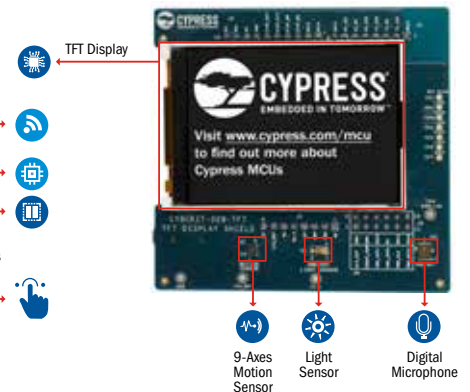
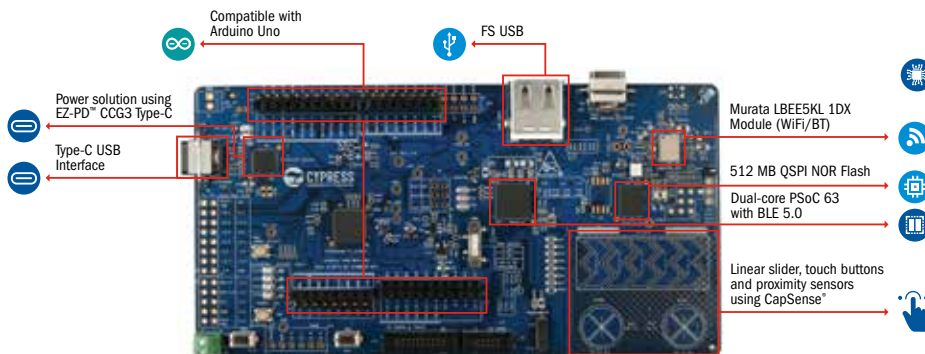
PSOC 6 BLE PROTOTYPING KIT
(CY8CPROTO-063-BLE)



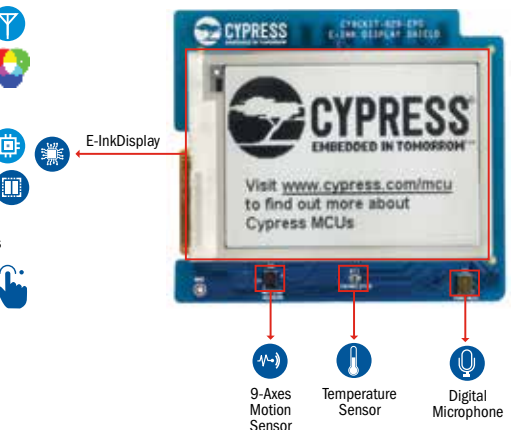
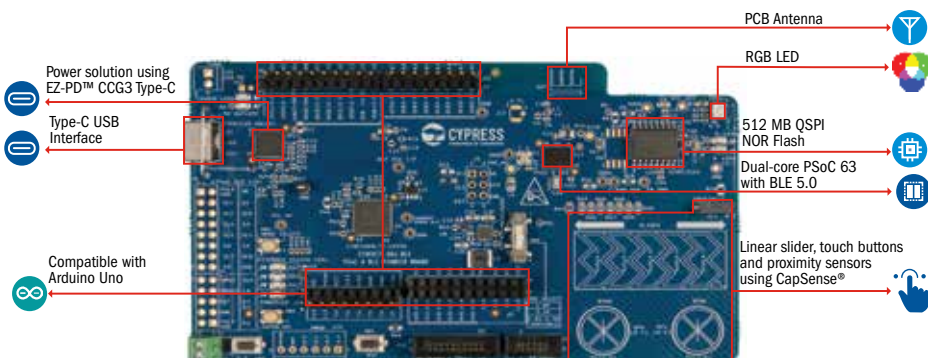
PSOC 64 SECURE BOOT PROTOTYPING KIT
(CY8CPROTO-064-SB)



PSOC 6 WIFI-BT PIONEER KIT
(CY8CKIT-062-WIFI-BT)



PSOC 6 BLE PIONEER KIT
(CY8CKIT-062-BLE)



For more information on PSoC 6 kits, go to:
<http://www.cypress.com/microcontrollers-mcus-kits>

SOLVE YOUR IoT PROBLEM HERE

What's your next game-changing application?



Graphic Display



Incorporate a slick graphics display to your next IoT application with the use of the Universal Digital Blocks (programmable logic).



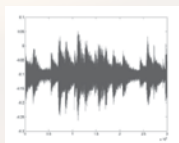
Audio Processing



Digital Microphone



Speaker



Signal Processing

Add an audio subsystem with an I²S interface and two Pulse Density Modulation (PDM) channels.



Sensor Fusion



Analog Sensors



Digital Sensors

Interface to analog sensors with the programmable analog block functions of PSoC 6, which includes op-amps, comparators, and a 12-bit SAR ADC. Digital sensors can easily be connected via the Serial Communication Blocks (SCBs).

PSoC 63 Connectivity Line

MCU Subsystem

arm

Cortex-M4
with SP FPU
150-MHz

I-Cache 8KB

arm

Cortex-M0+
100-MHz

I-Cache 8KB

SRAM
Up to 1MB

Flash: Up to 2MB

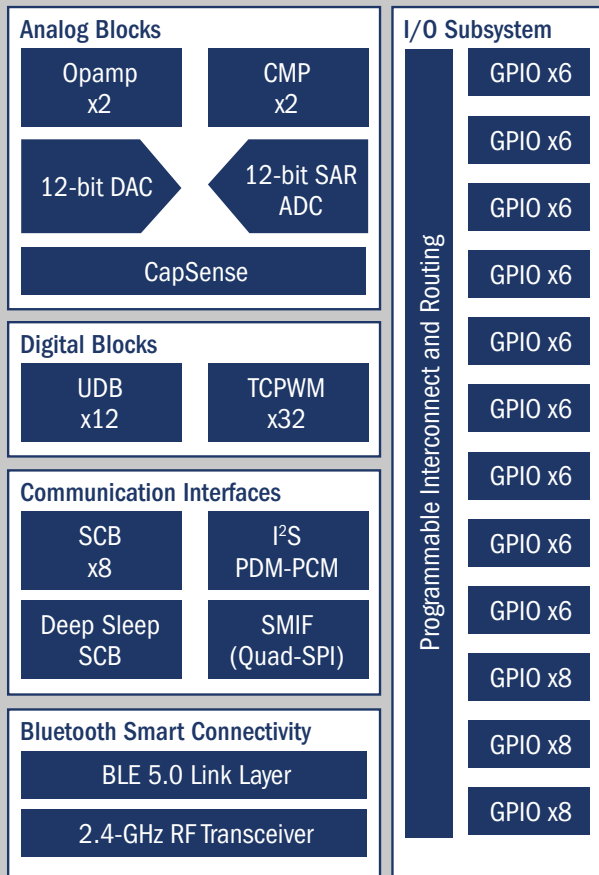
CRYPTO

DMA X2

eFUSE

RTC

Advanced Microcontroller Bus Architecture (AMBA-AHB) and IPC



Human Machine Interface



Sliders



Buttons



Proximity



Vibration Motor



Multiple LEDs

Replace traditional buttons with touch, add touch sliders, or add proximity detection with Cypress' CapSense technology. Use the TCPWM to drive motors, actuators or LEDs.



Interface Bridging and Port Expansion



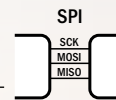
Full-speed USB Host or Device



UART



I²C BUS



SPI

Serial Interfaces

Serial Communication Blocks (SCBs) can be configured to support serial interfaces such as UART, I²C and SPI. USB 2.0 host and device support is available with PSoC 6.



Wireless Connectivity



Bluetooth



Wi-Fi

Add Bluetooth Low Energy connectivity with built-in BLE radio and sub-system. Add Wi-Fi or Bluetooth with Cypress' WICED portfolio.

PSOC®



LOWEST POWER,
MOST FLEXIBILITY,
BUILT-IN SECURITY.

BE A GAME-CHANGER. GET STARTED TODAY.

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ABOUT CYPRESS

Cypress is the leader in advanced embedded system solutions for the world's most-innovative automotive, industrial, home automation and appliances, consumer electronics and medical products. Cypress' programmable and general-purpose microcontrollers, analog ICs, wireless and USB-based connectivity solutions and reliable, high-performance memories help engineers design differentiated products and get them to market first. To learn more, go to www.cypress.com.



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[Cardiovascular Age](#)
[Cardio Capacity \(VO2 Max\)](#)
[Heart Rate Graph](#)
[Live Heart Rate](#)
[Troubleshooting Gaps in Heart Rate Graphs](#)

Heart Rate Graph

The Oura App provides the ability to monitor your heart rate 24 hours a day. The daytime heart rate graph is available for:

- [Gen3](#) and [Oura Ring 4](#) with **active membership**
- [iOS](#)
- [Android](#)

This feature is **not** available on Gen2 or older.

[What Is the Heart Rate Graph?](#)

[24-hour Heart Rate Trendline](#)
[Live Heart Rate](#)
[Restorative Time](#)
[Stress](#)
[Daytime Heart Rate Measurement](#)
[Activity Heart Rate](#)
[How to Use the Heart Rate Graph](#)
[Things to Keep in Mind](#)
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What Is the Heart Rate Graph?

The heart rate graph located on the home screen of the Oura App shows color-coded visualizations of your sleeping, daytime, stress, and activity heart rates, as well as periods of Restorative Time, all in the same chart.

The heart rate graph includes the following components:

24-hour Heart Rate Trendline

The 24-hour graph displays [sleep resting heart rate](#) in blue, [Restorative Time](#) in green, [daytime stress](#) in pink, and [Activity Heart Rate](#) in orange. The white trendline is the average of your heart rate readings throughout the 24-hour period.

Nighttime heart rate graphs are also available on the Sleep and Readiness screens of the Oura App.

Live Heart Rate

The latest reading of your pulse is displayed as beats per minute (BPM) beneath the heart rate graph, with a time indication of how long ago the last update was measured. Tapping the heart next to this number will update it with your current pulse measurement. Check out [the Live Heart Rate article](#) for more information.



Live Heart Rate

The latest reading of your pulse is displayed as beats per minute (BPM) beneath the heart rate graph, with a time indication of how long ago the last update was measured. Tapping the heart next to this number will update it with your current pulse measurement. Check out [the Live Heart Rate article](#) for more information.

Restorative Time

Restorative Time marks moments during the day when your body falls into a restful recovery state. Tracking these moments is a good way to measure your readiness. For more information on getting the most out of this feature, see [our Restorative Time article](#).

Stress

Moments where your body has felt high levels of physiological stress are marked in pink on the heart rate graph. For more information, see the [Daytime Stress article](#).

Daytime Heart Rate Measurement

Oura takes measurements of your daytime heart rate for one full minute, every five minutes, using the Oura Ring's green LEDs. To preserve battery and maximize accuracy, a daytime heart rate measurement is only taken under optimal conditions, which include low movement and balanced average body temperature.

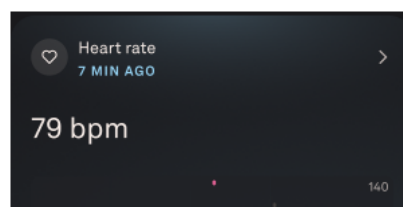
Due to prioritizing these conditions, you *may* not receive an automatically updated heart rate for up to 30 minutes, but it can be manually updated at any time by using [Live Heart Rate](#). Also note that "low movement" generally includes everyday actions, up to excessive motion associated with workouts.

Activity Heart Rate

The [Activity Heart Rate feature](#) is used to record your exercises. Heart rate measurements taken from physical activity are highlighted in orange on the graph.

How to Use the Heart Rate Graph

After your first full day of using the Oura Ring, the 24-hour heart rate graph displays as its own card on the Oura App's Today screen. The graph shows heart rate beats per minute (BPM) for the current day (midnight to midnight). The most recent heart rate value (in BPM) is displayed beneath the graph, along with the Live Heart Rate button.



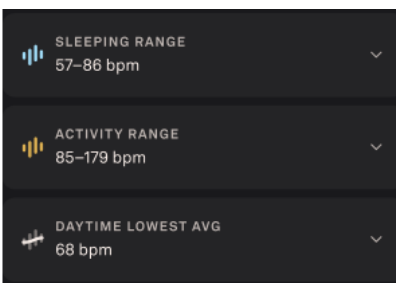


Each x-axis division of the graph represents a 30-minute time window, and every individual colored dot within those divisions is a five-minute heart rate range. Longer vertical bars indicate clusters of readings that overlap within the same 30 minutes.

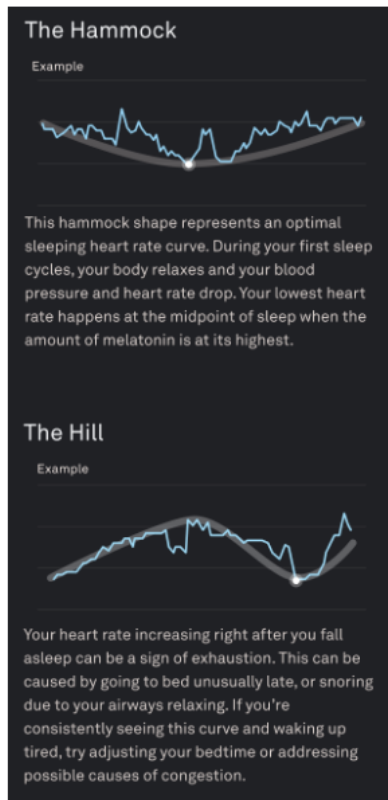
The solid white trendline across the graph represents the *average* of all heart rate measurement. Dotted white lines indicate periods in which reliable data is not available. See the [Troubleshooting Gaps in Heart Rate Graphs](#) article for more information.

Tap anywhere on the heart rate graph for more detail. (You can also access the Heart Rate detail screen through the Vitals tab or the shortcut at the top of the Today screen.) Press and hold or drag your finger on the graph to see a thin, vertical blue line. This line shows your *exact heart rate range* during that timeframe. Swipe right on the graph to view heart rate data from previous days.





Tap on any of the five core metrics (Restorative Time, Stress, Sleeping, Activity, and Daytime) to see more data. Select "Learn more" beneath each one for further information, including [how to interpret your heart rate curves](#).



Things to Keep in Mind

- Daytime Heart Rate and [Live Heart Rate](#) are not intended to record workouts. For accurate measurement of your pulse during exercise, see the [Activity Heart Rate](#) feature
- Excess movement or unusually low average body temperature may result in [gaps in your heart rate graph](#) data
- Confirmed naps appear as sleep (in blue) on the 24-hour daytime graph

More Information

[Troubleshooting Gaps in Heart Rate Graphs](#)

[Activity Heart Rate](#)

[Restorative Time](#)



Things to Keep in Mind

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[Activity Heart Rate](#)

[Restorative Time](#)

[Live Heart Rate](#)

[Daytime Heart Rate](#)

[Daytime Stress](#)

[Sleeping Heart Rate: Look for These 4 Patterns](#)

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Sleep Stages

Every night, your brain and body experience different phases of sleep, each essential for supporting your mental and physical health. With the Oura App, you can track the quantity and quality of your sleep, note your high points, and identify areas for improvement.

[What Are Sleep Stages?](#)

[How Sleep Stages Are Measured](#)

[Oura's Sleep Staging Algorithm](#)

[More Information](#)

What Are Sleep Stages?

In the Sleep detail view, you'll find color-coded sleep stage data at the top of the screen. Your [sleep stage hypnogram](#) is also available under the Details section.



The Oura Ring measures four separate stages of sleep:

- **Awake:** the time you spent in bed before and after falling asleep. This also includes [brief awakenings that occur during sleep](#)
- **REM:** [rapid-eye-movement \(REM\) sleep](#) is associated with dreaming, memory consolidation, and creativity. REM sleep is regulated by circadian rhythms and makes up anywhere from 5-50% of your total sleep time. Typically, you get more REM sleep during the latter half of the night
- **Light:** [light sleep](#) guides you into deeper stages, and acts as the beginning



includes [brief awakenings that occur during sleep](#)

- **REM:** [rapid-eye-movement \(REM\) sleep](#) is associated with dreaming, memory consolidation, and creativity. REM sleep is regulated by circadian rhythms and makes up anywhere from 5-50% of your total sleep time. Typically, you get more REM sleep during the latter half of the night
- **Light:** [light sleep](#) guides you into deeper stages, and acts as the beginning of or transition between each sleep cycle. Light sleep makes up about 50% of total sleep time
- **Deep:** also known as slow-wave sleep, [deep sleep](#) is when your body repairs and restores itself. During deep sleep, your muscles grow and repair, immune system is refreshed, and brain flushes out toxins. Deep sleep makes up anywhere from 0–35% of your total sleep. Usually, you get more deep sleep during the early night.

Every individual's sleep requirements are different, though the recommended amount is 7-9 hours for most adults. This is enough time for your body to go through four to five 90-minute sleep cycles. In general, you spend most of the night in light sleep.

Here are some general tips:

- Try to optimize for less awake time and more restorative deep or REM sleep. If you're getting more REM sleep than usual, this may be a sign you're recovering from prolonged sleep deprivation. Check out our blog for [tips on getting more REM sleep](#).
- If you're getting more deep sleep than normal, it could be due to a recent hard workout or changes to improve your sleep quality. See this blog article for [more ways to increase deep sleep](#).
- Remember, light sleep isn't "wasted sleep." Half of your total sleep time is usually made up of light sleep, which is a natural and healthy part of the sleep cycle. Learn more about [light sleep and its benefits](#) on the Pulse Blog.

How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

- Respiratory rate slows
- Heart rate decreases
- Average body temperature drops

Deep sleep:

- Low movement
- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:



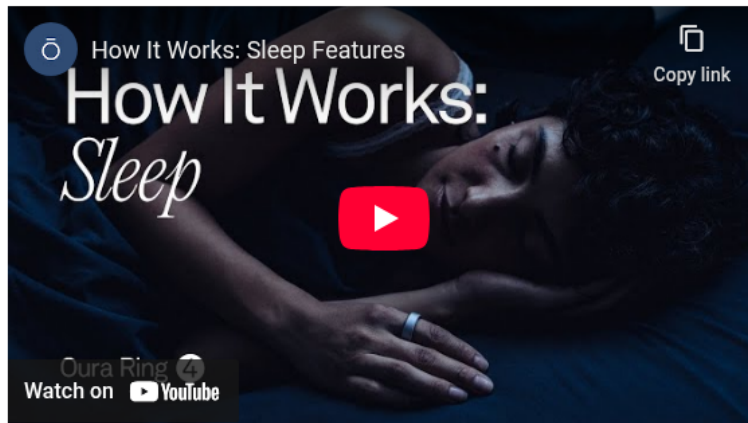
Deep sleep:

- Low movement
- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:



Oura's Sleep Staging Algorithm

Oura's sleep staging algorithm achieved 79% agreement with [polysomnography \(PSG\) sleep lab tests](#).

Find more information about these validation studies and the accuracy of our sleep algorithms on our blog: [Oura's New Sleep Staging Algorithm: More Accurate Than Ever Before](#).

You can export your sleep stage data from the Oura App to Apple Health through our [Apple Health integration](#).

More Information

[What Are the 4 Stages of Sleep?](#)

[Nighttime Movement Feature Displays Your Tossing & Turning](#)

[REM Sleep: What Is It and How to Get More](#)

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Nap Detection

The Nap Detection detection feature is available for:

- [Gen2](#)
- [Gen3](#) and [Oura Ring 4](#) with **active membership**
- [iOS](#)
- [Android](#)

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What is Nap Detection?

With Nap Detection, Oura users get credit for all accumulated sleep throughout the day and night. For those who regularly nap, or sleep more during the day due to shift-work, polyphasic sleep patterns, new parenthood, etc., Nap Detection will improve sleep tracking accuracy, and account for these sleep periods in Sleep and Readiness Scores.

If your nap is recorded after 6 PM, your Sleep and Readiness Scores won't be updated until the following morning. However, you'll still be able to see all sleep analysis data from your nap (i.e. sleep stages, RHR, HRV, etc). Once your scores update, they'll also be reflected in on the Sleep and Readiness screens, as well as in your [trends](#) graphs.

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one [sleep stage](#) (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the [24-hour heart rate graph](#) as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

Please note: In many cases, the Oura App won't detect a nap if it occurs within four hours of the end of the main sleep cycle. To increase the likelihood of



hour heart rate graph as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

Please note: In many cases, the Oura App won't detect a nap if it occurs within four hours of the end of the main sleep cycle. To increase the likelihood of additional sleep being included in your main sleep cycle, we recommend syncing your Oura Ring or placing it on the charger only **after your second wake-up time** (if your nap is within four hours of your first wake-up time).

If a nap at least 15 minutes long is detected, it will be added to the timeline on the Oura App's Today screen and to the Sleep screen. By tapping on the nap, you'll see full sleep analysis of your nap, including your hypnogram, amount of time spent in each stage, as well as your movement, resting heart rate, and heart rate variability graphs. If the nap is inaccurately recorded, you can delete it through this screen to remove its impact on your Sleep and Readiness Scores.

Your nap details will be shown in the app's home, Sleep, and Readiness screens.

Things to Keep in Mind

- The only Sleep and Readiness contributors *not* impacted by naps are Latency (included in your **Sleep Score**), Recovery Index, Activity, and Activity Balance (included in your **Readiness Score**)
- Your Sleep Score may be impacted (positively or negatively) depending on the quality of rest and timing of your nap
- Your Readiness Score may also be lowered if your average body temperature significantly elevates while you nap, as this can be a sign of emerging illness
- Nap Detection data is currently not available via Oura on the Web
- Naps cannot be manually adjusted

More Information

Sleep Stages

Sleep Contributors

Readiness Contributors

How Does the Oura Ring Track My Sleep?

Oura's Nap Detection Supports Diverse Sleeping Patterns

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Meet Oura

Oura's New Sleep Staging Algorithm: More Accurate Than Ever Before



Author: Oura Team | November 16, 2022



At Oura, we're committed to innovating and improving our health sensing algorithms through rigorous scientific processes, as well as sharing the research and validation with our community. Oura's Sleep Staging Algorithm 2.0, which matches performance with the gold standard [polysomnography \(PSG\) sleep lab test](#).

The result of more than two years of extensive research and development, this updated algorithm is based on a 2021 research study published in [Sensors](#) (a leading international, peer-reviewed, and open-access journal on the science and technology of sensors), which showcases the results of the algorithm.

How Accurately Does Oura Measure Sleep?

By leveraging every signal from the Oura Ring (movement, temperature, heart rate, and HRV), the [Sensors](#) study found Oura's sleep staging algorithm achieves **79% agreement** with gold-standard polysomnography (PSG) for classifying the [four stages of sleep](#) (wake, light, deep, and rapid eye movement (REM) sleep). By contrast, most commercial wrist-based wearables are limited to [60-65%](#)

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Oura's new sleep staging algorithm was developed using advanced machine learning techniques and one of the largest wearable sleep datasets collected to date. This one-of-a-kind dataset includes diverse participants across multiple continents and sleep laboratories, demonstrating the power of using all of the signals of the Oura Ring to maximize accuracy.

Read [the full paper in *Sensors*](#) and see key takeaways below.

Additional Validation Studies

Since then, additional research has been conducted on Oura's accuracy.

In 2024, a [study](#) conducted by the University of Tokyo further validated the accuracy of Oura Sleep Staging Algorithm 2.0. Comparing the accuracy of the sleep staging algorithm on Oura Ring Gen 3 to PSG, researchers found that Oura's measurements "did not significantly differ" from PSG, bringing further scientific credibility to Oura's sleep tracking ability. Read more about the key findings [here](#).

In 2023, the National University of Singapore published a [study](#) that compared six wearable sleep trackers. They found that Oura has better agreement with PSG than four other wearables, including Fitbit Sense. The only device with better performance was the EEG-based Dreem headband; however, one third of the study participants did not tolerate the headband and had to remove it to sleep.

A Global Approach

Oura has already collected more than 1,200 nights of data from participants across the globe





One of the main limitations of current academic research and commercial products is that sleep staging algorithms are typically developed on a limited amount of data (typically 20-30 nights) and homogenous populations. Because these algorithms are “trained” on a narrow dataset, they may not perform the same when they’re extended to a broader, more diverse population.

To counteract that limitation, Oura has developed an even more accurate sleep staging algorithm by actively building one of the largest sleep datasets to-date.

The dataset already contains over 1,200 nights (and counting!) of simultaneous PSG and Oura Ring data from sleep labs all over the world. It’s already enabled Oura’s scientists to train and evaluate the algorithm across a diverse set of individuals. Notably, the database captures diverse sleeping patterns, including physiological responses, across several factors:

- Different environments (e.g. weather, climate)
- Different skin tones
- Different body compositions (BMI)
- Unique behavioral factors (e.g. eating habits that could influence sleep, or different wakeup times and bedtimes)
- A wide age range (15-73) to account for sleep staging changes that come with age (e.g., deep sleep decreases with age)
- People with and without sleep disorders

Using Every Metric

Most wearables use a combination of different body signals (e.g. heart rate, movement, etc.) to determine when you’ve fallen asleep and which sleep stage you’re currently in. Each sleep stage shows a distinctive pattern. For example, during deep sleep, your heart rate decreases, your respiration stabilizes (like a metronome), and your body’s temperature drops.

So, if you want to get a clear look at what’s happening on the inside after you’ve fallen asleep, the quality and accuracy of your sleep insights will depend upon your device’s ability to correctly measure and identify these distinct patterns.

Oura’s unique ability to leverage movement, temperature, heart rate, heart rate variability (HRV), and mathematical modeling that incorporates well known sleep patterns (e.g, REM sleep tends to occur later in the night) results in state-of-the-art sleep staging accuracy. The example below shows how closely Oura is able to

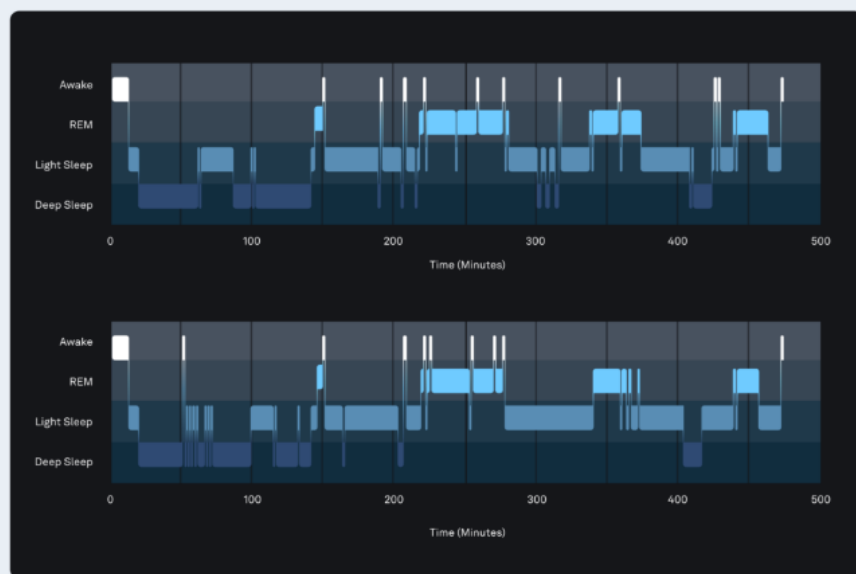


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LEARN MORE: [How Does the Oura Ring Track My Sleep?](#)

Below is an example hypnogram for a representative night of the validation dataset.



The top graph shows the sleep stages across the night determined with gold standard polysomnography. The bottom graph shows the sleep stages predicted by the Oura Ring algorithm (accuracy = 84%).

Read [the full paper in Sensors](#) for more insights on the relation between temperature, HRV, and different sleep stages.

Accuracy Across Sleep Stages

One of the main strengths of this algorithm is the high sensitivity and specificity it achieves across all sleep stages — ranging from 74% to 98%. While other studies have shown similar results for the detection of a *specific* sleep stage, this improved performance typically comes at the expense of the others (e.g. high performance in detecting deep sleep might result in a poor ability to detect REM).

This research demonstrates that a sophisticated model can maintain high sensitivity and specificity for *all sleep stages*.

Oura maintains high accuracy, sensitivity, and specificity across all sleep stages.



This research demonstrates that a sophisticated model can maintain high sensitivity and specificity for *all sleep stages*.

Oura maintains high accuracy, sensitivity, and specificity across all sleep stages.

	Accuracy	Sensitivity	Specificity
Awake	96%	81%	98%
REM	91%	78%	95%
Light Sleep	80%	82%	79%
Deep Sleep	91%	74%	95%

Read [the full paper in Sensors](#) for additional box plots and graphs.

From the Oura Research Team

Lead author, Marco Altini, PhD, shares his excitement about sharing the new sleep staging algorithm with the Oura community:

“To develop the new sleep staging algorithm, we’ve collected a large dataset across the world, built key features relying on high-quality data from the sensors in the Oura Ring, and implemented a machine learning model outperforming current state of the art tools. For the first time, sleep staging using wearables not relying on EEG data, moved from 60–65% accuracy, up to about 80%. In the past few months, engineers and data scientists at Oura have taken our model and embedded it in the Oura Ring, so that every user can now benefit from the improved sleep stage detection accuracy. These are exciting times, as people will be able to better understand the impact of lifestyle and other factors on their sleep, while scientists worldwide will have an additional tool at their disposal to study sleep in real life settings, outside of the lab.”

In the meantime, we will continue to work directly with experts to advance sleep research and, of course, keep the Oura community updated each step of the way. Below, a note from Shyamal Patel, PhD, Head of Science at Oura:

“Our commitment to scientific validation and accuracy is exemplified in Oura’s research and development approach for the new sleep staging algorithm. This algorithm represents a significant leap in sleep tracking with consumer wearables and a major milestone for Oura. Above all, we are excited to put this algorithm into the hands of Oura members and empower them to measure, understand and improve their sleep.”



Responses From the Research Community

“In a world where wearable sleep trackers have been commoditized, it is great that one company has taken the trouble not only to validate its products across multiple age groups but also reveal to sleep professionals the methodology underlying their sleep stage estimation. Oura stands out as being dedicated to proving and improving its craft. My

"In a world where wearable sleep trackers have been commoditized, it is great that one company has taken the trouble not only to validate its products across multiple age groups but also reveal to sleep professionals the methodology underlying their sleep stage estimation. Oura stands out as being dedicated to proving and improving its craft. My team and I are delighted to be part of their transformation of sleep, health and wellbeing."

— Michael Chee, Professor and Director, Centre for Sleep and Cognition, Yong Loo Lin School of Medicine, National University of Singapore.

"This most recent paper evaluating the performance of the Oura Ring in measuring sleep is a wonderful example of what can be achieved when deep knowledge of sleep science is combined with innovative technology and state of the art modeling techniques. The combination of the technical performance and extremely usable form factor makes the Oura Ring a potential savior for the problem of how to ubiquitously measure sleep in large numbers of people. It represents a real breakthrough for the field of sleep science and potentially for sleep medicine."

— Ian Colrain, President, SRI Biosciences and Professorial Fellow, School of Psychological Sciences, The University of Melbourne

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